SAMPLE

Earth's Sky: The Sky From Other Worlds

What the Sky Looks Like on Mars, the Moon, and Titan

Aira Astralis

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Introduction

Stand on the surface of another world and the first thing that changes is the sky. The Sun may hang three times too large, or shrink to a brilliant star; the daylight may be white, or butterscotch, or a thick twilight orange that never quite ends; the Sun may climb out of the western horizon and set in the eastern sky, or not rise at all for a season, or never rise at all because the world is locked with one face forever sunward. Above the horizon, the Moon may be missing, or multiplied into a dozen moons that rise and set like slow lanterns, or fixed forever in one place like a monument. This book is about those skies: the sky as it would be seen, measured, and experienced from the surfaces and cloud decks of the Sun's other worlds, from airless Mercury out to the double-locked pair of Pluto and Charon, and beyond to the planets of other stars, where suns come in pairs and sunsets can be double.

The subject is not fantasy. Every sky described in this book is grounded in measured fact: the known atmospheres, rotation states, orbital distances, and moon systems of the planets, the direct photographs of Martian sunsets and Titan's haze, the calculated brightness of the Sun at each distance. Astronomers and spacecraft have been assembling the sky views of other worlds for more than half a century, and the pieces are good enough to reconstruct. Some of these skies have been photographed from the ground by landers; some have been modeled from orbit with exquisite precision; some can only be computed, but computed from hard numbers, not imagination.

The tour follows the family order this series always uses. The opening chapters cover the discovery of the subject, the long human effort to picture the skies of other worlds, and the basic physics that explains why a sky has a color, a brightness, and a daily cycle at all. The middle chapters walk the skies themselves, world by world, and then set them in the wider context of spaceflight and observation. The later chapters cover the measurements that pin the reconstructions down, the significant discoveries, the open questions, the coming missions and instruments, and the trivia that makes the subject a pleasure to keep learning. By the end, the sky over your head should feel like one member of a very large family, a local example of a phenomenon the universe produces in endless variety, and you should be able to work out, from a few facts about any world, roughly what its sky would do to an observer standing under it.

The chapters also keep a running account of how the knowledge was won, because the evidence is half the pleasure: a sky reconstructed from a lander's photograph carries a different authority than one computed from a flyby, and this book marks the difference wherever it matters. One promise, and one warning. The promise: everything asserted here rests on the published record of planetary science, from the first telescopic hints to the landing cameras, and the speculative corners are labeled as such. The warning: a sky is not a still picture. Every sky in this book moves, and the motion is half the story, the length of the day, the path of the Sun, the parades of moons and the slow wheel of the seasons. Read the tour as an invitation to imagine standing still on a moving world, and the sky will do the rest.

Chapter 1: The Discovery of The Sky From Other Worlds

The discovery that other worlds have skies happened in stages, and the first stage was the discovery that other worlds are worlds at all. To the naked-eye astronomers of antiquity, the planets were points of light with personalities; the Moon alone showed a face, and its face was read as weather signs and terrain at once, a man, a rabbit, a set of seas, by cultures that had no way to check. The idea that the Moon might be a place, with its own ground and its own sky above that ground, had to wait for the telescope, and it came fast once the telescope arrived. Galileo, turning his rough optick on the Moon in 1609 and 1610, saw mountains with measured shadows, and understood at once that the lunar world was landscape, not emblem: a territory with a geometry like Earth's, lit by the same Sun. He did not yet know that the sky over that landscape would be black at noon, but the question had been opened and could never be closed again.

The same Galileo found Jupiter's four large moons, and with them the first proof that other worlds can have their own monthly spectacles. A Jovian observer, watching the inner moon Io cross the sky, would see it flare into a small brilliant point and then dim into shadow as it entered Jupiter's cone of darkness; the spectacle was visible in miniature from Earth, as a measured wink in Galileo's records, and the wink was enough to establish that other worlds come complete with their own eclipses, their own month, their own sky traffic. The moons of Jupiter were also, historically, the first clock in the sky: their regular eclipse timings were used to measure the speed of light, when astronomers noticed the timings running late when Jupiter was far. Other-world astronomy was doing terrestrial physics before anyone had ever spoken the phrase.

Mars taught the next lesson, less happily. The planet's polar caps and seasonal changes were real discoveries of the telescope era, but the famous canals were a misreading, a network of faint linear markings reported from the 1870s onward and built by some observers into a civilization's irrigation system. The canal era mattered to this book's subject for what it reveals: the era's greatest observers were already trying to see another world as a place someone could stand, with weather, seasons, and a sky worth looking up at. The misreading was corrected by the first flybys, when Mariner 4 arrived in 1965 and returned a cratered, airless-looking surface, and the correction was itself corrected when later orbiters found valleys and river channels. The true Martian sky, butterscotch at noon and blue around the setting Sun, was still ahead, waiting for cameras on the ground. The first successful planetary flyby belongs to this stage of the story too: Mariner 2, crossing Venus in December 1962, measured the planet's radiometric temperature from flyby range and found the surface hot far beyond the boiling point of water, the first direct evidence that the veiled world's clouds hid a furnace rather than a swamp, and the first data point in the long reconstruction of what light, if any, survives the trip to that ground.

The ground came in 1976. The Viking landers, the first successful spacecraft to sit on another planet and photograph from its surface, returned panoramas of a rock-strewn plain under a pale grayish-pink sky, and the pictures unsettled expectations in both directions: the sky was lighter than predicted, because the fine dust scatters light far more than thin air alone would, and it was stranger than predicted, because that dust paints the Sun's neighborhood blue at the day's edges. For the first time, the sky of another world was not an inference or a rendering. It was a photograph, taken from the ground, with a horizon in it and everything.

Then the sky became routine. The Soviet Venera landers, hardened to survive the Venusian furnace, sent back the first pictures taken from under another planet's clouds, dim orange scenes lit like a deeply overcast day, and measured the light that filters down: the Venusian surface receives a few percent of the sunlight that strikes the cloud tops, diffused and reddened on the way. Voyager flew the outer planets and found each system of moons a new set of skies: the sulfur-orange clouds of Io with Jupiter swollen across their heaven, the methane-blue of Uranus and Neptune from orbit, and Titan, the moon with the thickest atmosphere in the solar system, waiting to be landed on. Cassini orbited Saturn for thirteen years and watched the rings' shadow crawl across the planet's clouds in slow seasons; its Huygens probe parachuted onto Titan in 2005 and photographed the ground-level view under the orange haze, a landscape of ice pebbles carved by methane rain, lit by a Sun rendered a diffuse glow.

The most recent entries in the discovery record are the far and the small. The New Horizons flyby of 2015 resolved Pluto from a smudge into a world with a bright heart-shaped plain and a sky thin enough to haze blue at the horizon; the same craft, looking back sunward, photographed Pluto's thin blue limb against the dark beyond it. And the exoplanet era has opened skies no probe will reach soon: the first directly imaged planets of other stars, the measured orbits of circumbinary planets around double suns, the transit spectra that read the upper atmospheres of alien worlds by the light they swallow. The skies of other worlds, once a philosophical curiosity, are now a working research frontier, and every mission that lands anywhere adds a new photograph to the album.

The speculative tradition between the telescope and the spacecraft deserves a paragraph of respect, because it kept the subject alive when nothing could be checked. Christiaan Huygens, in his Cosmotheoros of 1698, reasoned carefully about what Jupiter's moons and Jupiter itself would look like to an observer among them, and about the inhabitants his era assumed must live there, and his arithmetic of sizes and distances, if not his biology, was sound; the book became the century's bestseller on the subject and established the habit of asking the observer's-eye question about other worlds. The habit survived in the planetary literature even after the canals died: the mid-twentieth-century textbooks routinely included a paragraph on the color of the Martian sky, arguing from the thinness of the air and the loading of the dust to predictions that ranged from near-black through deep blue to the pale butterscotch the landers eventually found, and the spread of the predictions, each defensible, is a standing lesson in how underdetermined a sky is until someone stands under it.

The Apollo program contributed the one sky every schoolchild now recognizes, the black lunar noon, and the photographs carried a small controversy inside them: the eye's adaptation fooled early readers of the images into reading a bright gray sky where the instruments recorded black, because the mind refuses to attach hard sunlight to a star-filled background, and the calibration targets the astronauts deployed exist partly to settle exactly such arguments. The same program, on Apollo 15, left a camera pointed at the lunar horizon long enough to catch the glow of the solar corona in the minutes of a deliberate darkening, one of the very few sky observations that requires no atmosphere at all, and a reminder that even the skyless sky holds phenomena worth an astronaut's scarce time.